

Homework 2

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2022-11-21

Deadline: 2022-11-27 22:30:00, Beijing time

How to submit?

Email: YJ40202@163.com

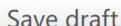
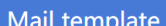
Subject: [homework](#)

Attachments:

Armstrong.java, LargestCommonPrefix.java, Palindrome.java and Emirp.java

Note: Don't use package and don't submit compressed file.

How to submit?

 Send  Save draft  Cancel |  Mail template  Mass Mailing

Recipient : YJ40202<yj40202@163.com>  email address

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 Attach files  Cloud attachment | (100%, 2.72K uploaded, total file size2.72K)

 Divisible.java 610B Uploaded	 Delete	 Eqqs.java 915B Uploaded	 Delete	 Temperature.java 581B Uploaded
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Armstrong.java

An Armstrong number is an n-digit integer such that the sum of the nth power of its digits is equal to the number itself. For example, 371 is an Armstrong number because $3^3 + 7^3 + 1^3 = 371$ (371 is a 3-digit number). 8208 is an Armstrong number because $8^4 + 2^4 + 0^4 + 8^4 = 8208$ (8208 is a 4-digit number).

Write a program that asks the user for a lower limit and an upper limit and prints all the Armstrong numbers up to that limit.

You should use the following method header in your program.

```
/** you need write java docs here */  
public static boolean isArmstrong(int number)
```

Armstrong.java

Here are some sample runs:

```
Enter the lower limit: 10 [Enter]
Enter the upper limit: 1000 [Enter]
The Armstrong numbers between 10 and 1000 are: 153 370 371 407
```

```
Enter the lower limit: 8000 [Enter]
Enter the upper limit: 20000 [Enter]
The Armstrong numbers between 8000 and 20000 are: 8208 9474
```

```
Enter the lower limit: 25000 [Enter]
Enter the upper limit: 100000 [Enter]
The Armstrong numbers between 25000 and 100000 are: 54748 92727 93084
```

LargestCommonPrefix.java

Write a program that prompts the user to enter two strings and display the largest common prefix of the two strings. If there are no common prefix between the two entered strings display a message which tells the user that the two string doesn't have a common prefix.

LargestCommonPrefix.java

Here are some sample runs:

```
Enter the first string: Atlanta [Enter]
Enter the second string: Macon [Enter]
Atlanta and Macon have no common prefix.
```

```
Enter the first string: Welcome to Java [Enter]
Enter the second string: Welcome to programming [Enter]
The common prefix is: Welcome to
```

```
Enter the first string: I love coffee [Enter]
Enter the second string: I love Java [Enter]
The common prefix is I love
```

Palindrome.java

Write a method that checks whether the input string or a sentence (a string with spaces) is a palindrome or not. The method should be case insensitive and should ignore spaces. Write a test program that prompts the user to input a string and invokes this method.

You should use the following method header in your program.

```
/** you need write java docs here */  
public static boolean isPalindrome(String str)
```


Palindrome.java

Here are some sample runs:

```
Enter a string: madam [Enter]
madam is a palindrome
```

```
Enter a string: banana [Enter]
banana is not a palindrome
```

```
Enter a string: Race      Car [Enter]
Race      Car is a palindrome
```

```
Enter a string: Too      HOT to hoot [Enter]
Too      HOT to hoot      is a palindrome
```

Emirp.java

An emirp (prime spelled backward) is a non-palindromic prime number whose reversal is also a prime. For example, 17 is a prime and 71 is a prime, so 17 and 71 are emirps. Write a program that displays the first 100 emirps. Display 10 numbers per line, separated by exactly one space, as follows:

```
13 17 31 37 71 73 79 97 107 113
149 157 167 179 199 311 337 347 359 389
...
```

You should use the following method headers

```
/** write some java here */
static boolean isPalindrome(int number)
/** write some java here */
public static boolean isPrime(int num)
/** Reverse the given number, for example, return 321 if number is 123 */
static int reverse(int number)
```